

Estimating the germination dynamics of *Plasmopara viticola* oospores using hydro-thermal time

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The effects of environmental conditions on the variability in germination dynamics of *Plasmopara viticola* oospores were studied from 1999 to 2003. The germination course was determined indirectly as the relative infection incidence (RII) occurring on grape leaf discs kept in contact with oospores sampled from a vineyard between March and July. The time elapsed between 1 January and the infection occurrence was expressed as physiological time, using four methods: (i) sums of daily temperatures $> 8^{\circ}\text{C}$; (ii) hourly temperatures $> 10^{\circ}\text{C}$; (iii) sums of hourly rates from a temperature-dependent function; or (iv) sums of these rates in hours with a rain or vapour pressure deficit ≤ 4.5 hPa (hydro-thermal time, HT). An equation of Gompertz in the form $\text{RII} = \exp[-a \cdot \exp(-b \cdot \text{HT})]$ produced an accurate fit for both separate years ($R^2 = 0.97$ to 0.99) and pooled data ($R^2 = 0.89$), as well as a good accuracy in cross-estimating new data (r between observed and cross-estimated data were between 0.93 and 0.99 , $P < 0.0001$). It also accounted for a great part of the variability in oospore germination between years and both between and within sampling periods. Therefore, the equation of Gompertz (with $a = 15.9 \pm 2.63$ and $b = 0.653 \pm 0.034$) calculated over hydro-thermal time, a physiological time accounting for the effects of both temperature and moisture, produced a consistent modelling of the general relationships between the germination dynamics of a population of *P. viticola* oospores and weather conditions. It represents the relative density of the seasonal oospores that should have produced sporangia when they have experienced favourable conditions for germination.

Keywords: environmental conditions, grapevine downy mildew, modelling, primary inoculum, *Vitis vinifera*

Introduction

Plasmopara viticola is the causal agent of grapevine downy mildew, an important disease in all grape-growing areas characterized by temperate climate and frequent rain during spring and summer (Lafon & Clerjeau, 1988).

Oospores represent the sexual stage of the pathogen, which overwinter in the leaf litter on the vineyard ground or buried in soil. Oospores are the sole source of inoculum for primary downy mildew infections in the next season (Zachos, 1959): in spring, they germinate into a sporangium that releases zoospores, which are responsible for infections on grape leaves and clusters (Galbiati & Longhin, 1984). Five to 18 days after infection has occurred, depending on temperature, the oomycete produces sporangia containing asexually-produced zoospores (Lalancette *et al.*, 1988b) that cause repeated

secondary infections under suitable environmental conditions (Lalancette *et al.*, 1988a).

In European areas with high humidity, the disease is potentially destructive and requires protection with fungicides; in Northern Italy, for instance, 7 to 10 fungicide treatments are usually applied against the disease. Some of these sprays, however, are applied as insurance against the highly erratic appearance of the disease and the damage it causes. To limit applications of unnecessary sprays, several weather-driven epidemiological models have been elaborated, but they often fail in predicting the real development of epidemics, which restricts their use in practice (Vercesi & Liberati, 2001).

According to the traditional conception of pathogen epidemiology, most of these models are based on the assumption that an epidemic starts from a restricted number of germinating oospores, so that the explosive increase of the epidemic is caused by a massive clonal multiplication causing secondary infections (Lafon & Clerjeau, 1988). Recent studies carried out with polymorphic microsatellite markers for *P. viticola* demonstrated

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that there is a continuous input of new genotypes into the epidemic during a prolonged period, each of them contributing one or a few lesions to the total disease severity, and only one or two genotypes per epidemic underwent secondary cycles and generated a high number of progeny (Gobbin *et al.*, 2005). Therefore, oospores play a key role in the development of downy mildew epidemics.

Current knowledge indicates that models for dimorphic pathogens such as *P. viticola* can be significantly improved only with quantitative modelling of both sexual and asexual phases (Rossi *et al.*, 2007). Although germination dynamics of oospores have been extensively studied there are some inconsistent results: (i) germination is different year to year and shows irregular fluctuations between oospore samples collected at different times of the same season, as well as within the same sample (Serra & Borgo, 1995; Hill, 2000; Pertot & Zulini, 2003); (ii) these fluctuations are not convincingly explained by high variability within oospore populations, unspecified edaphic factors or changing meteorological conditions (Burruano *et al.*, 1989; Vercesi *et al.*, 2000).

The aim of this work was to study variability in germination dynamics of *P. viticola* oospores based on the effects of environmental conditions. For this purpose, four methods for calculating the physiological time required for oospores to germinate were compared for their ability to fit actual data on oospore germination collected over a 5-year period.

Material and methods

Oospore samplings and germination tests

Oospore germination was determined for a 5-year period (1999 to 2003) in an experimental plot (about 500 m² wide) within a cv. Barbera vineyard located at Asti (Piedmont, Italy), which is usually affected by severe downy mildew epidemics. Fungicides against *P. viticola* were applied after the first seasonal onset of the disease, following the common practice, to ensure a representative overwintering inoculum.

Oospore germination was determined indirectly by testing for their ability to cause infection on grape leaf discs kept in contact with them under optimal laboratory conditions (Hill, 1998).

At the end of each grape-growing season (1998 to 2002), leaves showing typical and severe disease symptoms were collected before leaf fall. Leaves were dried in a jute bag at room temperature for 15 days, crumbled and located in the vineyard. Leaf fragments were put into a nylon mesh bag and positioned in plastic boxes (50 × 30 × 10 cm) with a sand layer on the bottom. The plastic boxes were buried in soil, such that the leaf fragments were placed at ground level, and left uncovered so that leaf fragments experienced natural weather conditions.

Between late March and early July, 10 g samples of leaf fragments were collected from the box once a week. In the laboratory, they were suspended in 50 mL of sterile water

and placed in Petri dishes (90 mm in diameter, three replicate plates per sample) which were incubated in a room at 20 ± 2°C for 14 days under natural light. Leaf discs (20 mm in diameter) excised from potted plants of cv. Barbera grown in a greenhouse and isolated from any external source of *P. viticola* inoculum were floated in the oospore suspension (20 leaf discs per plate) for 24 h, abaxial side down. After a 24-h period of exposure to oospores, the leaf discs were removed and replaced with new discs. The removed discs were placed on a layer of wet blotting paper in Petri dishes, with the abaxial surface exposed, and incubated at 20°C, 12 h light, 12 h dark. They were inspected daily for 10 days to detect the appearance of disease signs such as sporangiophores. A total of 280 leaf discs were considered per sample (20 discs per day × 14 days).

The relative incidence of infection (RII) was calculated for each sampling date and for each 24-h incubation period by dividing the number of leaf discs showing infection by the total number of discs infected over the entire sampling season. For example, 23 leaf discs showed downy mildew symptoms after contact with the oospores collected on 22 May 2000: 12, 9 and 2 discs that remained in contact for 24 h with the oospores dipped in water for 24, 48 and 72 h, respectively. Since a total of 276 leaf discs were infected in the 2000 sampling season, RII was 0.043, 0.033 and 0.007 for the three exposures, respectively.

Weather data and field observations

Meteorological data were collected in the vineyard by an automatic station (SIAP, Bologna, Italy) with a step of one hour.

The vineyard was carefully inspected at least once a week starting from bud burst to determine the time of the first seasonal appearance of disease symptoms such as 'oil spots' on leaves or affected bunches. The corresponding periods when the infections had probably taken place were determined following Rossi *et al.* (2002).

Seasonal patterns of oospore germination

Patterns of RII over the sampling season were studied by expressing the time in physiological units (Lowell *et al.*, 2004) calculated by four different methods.

The first two methods are sums of the daily temperatures (T) exceeding 8°C (degree-days, DD) and of the hourly T greater than 10°C (degree-hours, DH) starting from 1 January. The other two methods refer to the rate summation method (Curry & Feldman, 1987) and consist of sums of hourly rates from a temperature-dependent function that accounts for the effects of either T (thermal time, TT) or T and moisture (hydro-thermal time, HT) on oospore development.

These variables were calculated as:

$$DD_i = \sum_{i=1}^n (T_i - 8); \text{ when } T_i < 8, T_i = 8 \quad (1)$$

$$DH_i = \sum_{i=1}^n (T_i - 10); \text{ when } T_i < 10, T_i = 10 \quad (2)$$

$$TT_i = \sum_{i=1}^n 1/(1330.1 - 116.19 \cdot T_i + 2.6256 \cdot T_i^2);$$

when $T_i < 0$, $T_i = 0$ (3)

$$HT_i = \sum_{i=1}^n (TT_i \cdot M_i) \quad (4)$$

where i is the subscript for days in equation (1) and for hours in (2) to (4), with $i = 1$ (on 1 January, 01:00 hours) to n , T_i is the air temperature in each i th time unit, i.e. day or hour (daily T was calculated as a mean of 24 hourly values, 01:00 hours to 24:00 hours), and M is a dichotomic variable accounting for the moisture of the leaf litter holding oospores, as follows:

when $R_i > 0$ mm or $VPD_i \leq 4.5$ hPa, $M_i = 1$

when $R_i = 0$ mm and $VPD_i > 4.5$ hPa, $M_i = 0$

where R is rainfall (in mm) and VPD is the vapour pressure deficit (in hPa) calculated from temperature and relative humidity (RH, in %) following Buck (1981):

$$VPD = (1 - RH/100) \cdot 6.11 \cdot \exp[(17.47 \cdot T)/(239 + T)]$$

Calculations began on 1 January, 01:00 hours, of each year using the meteorological data measured in the vineyard until each sampling date, 10:00 hours, and then using the conditions of the laboratory test for the time that oospores remained in these conditions, i.e. $T_i = 20^\circ\text{C}$ and $M_i = 1$.

The effect of temperature in equation [3] was derived from Laviola *et al.* (1986). These authors incubated oospores at constant temperatures (5 to 35°C), dipped in water, and measured both the time required for oospores to germinate and germination frequency. Data on the minimum time (t in hours) for germination to occur were fitted against temperature with the following polynomial equation: $t = 1330.1 - 116.19 \cdot T + 2.6256 \cdot T^2$ ($R^2 = 0.98$, $SE_{\text{est}} = 0.765$). The ratio $1/t$ was then considered in equation (3) as the progress rate of germination determined by any temperature.

The effect of moisture in equation [4] was derived from Rossi & Caffi (2007). These authors incubated oospores at different regimes of water activity (a_w) for different times, at the optimum temperature of 20°C . They found a significant shift in germination when oospores were incubated at $a_w < 0.56$. The same authors also studied patterns of moisture in grape leaves forming the leaf litter holding the overwintering oospores in relation to weather conditions, and they found a close relationship between VPD and a_w , so that when $a_w = 0.56$, $VPD = 2.13$ hPa, with an upper 90% confidence limit of 4.5 hPa.

Using equations (1) to (4), physiological time was calculated for each leaf disc sample that showed infection in the laboratory test; the corresponding values of RII were cumulated over such a time and then regressed against DD, DH, TT or HT. In a preliminary analysis, the following equations were used: logistic, monomolecular and Gompertz in the forms shown by Campbell & Madden (1990). The equation parameters were estimated using the nonlinear regression procedure of SPSS (ver. 13, SPSS Inc.), which minimises the residual sums of squares using the Marquardt algorithm. The magnitude of the standard

errors of the model parameters, R^2 adjusted for the degree of freedom, the number of iterations taken by the Marquardt algorithm to converge on parameter estimates, and the magnitude and distribution of residues were considered to evaluate the goodness of fit since the best fits were obtained using the equation of Gompertz in the form: $y = \exp[-a \cdot \exp(-b \cdot x)]$; where a is the equation parameter accounting for the lag of the disease progress curve, and b is the rate parameter. Results of the other equations are not shown in this work.

The four methods were compared to assess their ability to reduce the between-year variability in estimating RII dynamics. For this purpose, regression equations were calculated for each year and the values of DD, DH, TT and HT (x in the Gompertz equations) were determined when RII (y) was set at 0.05 (i.e. 5% of seasonal infections), 0.1, 0.37 (i.e. the inflection point of the Gompertz curve) and 0.9, using the following equation: $x = (-1/b) \cdot \ln\{-[\ln(y)/a]\}$. Mean and standard deviation (sd) of x in the five years were then calculated and the four methods were compared using the coefficient of variation (CV), as: $(\text{sd}/\text{mean}) \cdot 100$ (Clewer & Scarisbrick, 2001). High CV values indicate a high between years variability in reaching the fixed value of x .

The four methods were also compared for their ability to estimate RII dynamics for a different year using a cross-validation technique (Jones & Carberry, 1994). Five regression equations were calculated using the pooled data of four years, excluding one year in turn; afterwards, these equations were used to estimate the data for the year not included. These cross-estimated data were compared with the observed data by calculating the Pearson's coefficients of correlation and the differences 'observed minus cross-estimated' for each year. Estimates were also regressed against observed data using the equation $y = a + b \cdot x$, and the null hypothesis that $a = 0$ and $b = 1$ was tested following Teng (1981).

Results

Oospore germination in laboratory tests

The germination course of oospores and the consequent infection of the grape leaf discs kept in contact with them showed high variability between years, between sampling times within a year and within each sample (Fig. 1).

In the oospore samples collected in the early season there was no germination, as in 1999 (Fig. 1a), or germination occurred after long periods of incubation in water, as in 2000 (Fig. 1b), 2002 (Fig. 1d) and particularly in 2003, when this situation extended until late April (Fig. 1e). Nevertheless, in 2001, oospores germinated in a short time in the early season (Fig. 1c).

In the oospore samples collected in the mid-season, germination usually began after a short time in water. In some samples, oospore germination was concentrated within a few days, as, for example, in most of samples collected in 2002 (Fig. 1d) and 2003 (Fig. 1e); however, in other samples, germination was distributed over a long

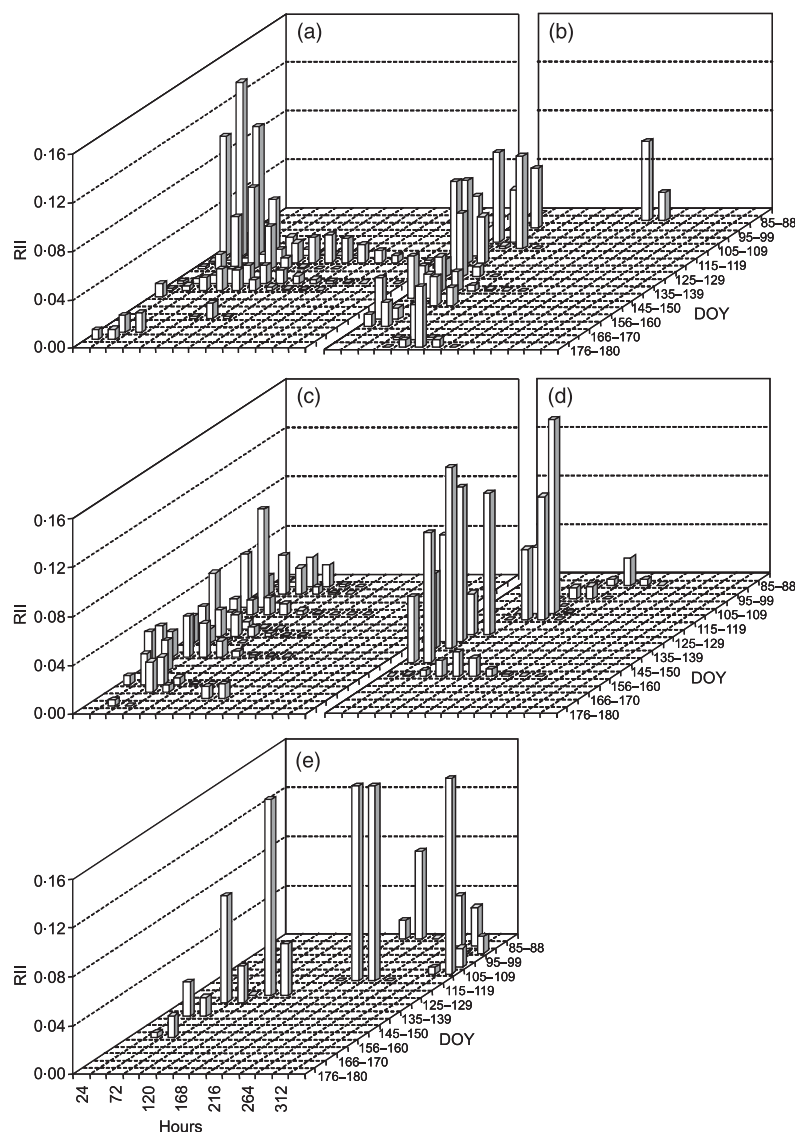


Figure 1 Distribution over time when *Plasmopara viticola* oospores were sampled (DOY, day of the year) and over time of incubation in water (hours) of the relative infection incidence (RII) in grape leaf discs kept in contact with the oospores under optimal conditions, yearly from 1999 (a) to 2003 (e). RII was calculated for each sample and incubation period by dividing the number of leaf discs showing infection by the total number of discs infected over the entire season (87, 276, 272, 125 and 91 infected discs in years 1999–2003, respectively).

period of incubation in water, as in many samples from 1999 (Fig. 1a) and 2001 (Fig. 1c).

In the oospore samples collected in the late season, germination became irregular, being delayed in some samples, such as in 1999 (Fig. 1a) and 2001 (Fig. 1c), or absent in others, such as in 2002 (Fig. 1d) and 2003 (Fig. 1e).

Germination dynamics over the physiological times

The Gompertz equation produced an accurate fit of the within-year relationships between relative infection by *P. viticola* in the grape leaf discs kept in contact with oospores and the physiological time expressed as DD, DH, TT or HT (Table 1). The R^2 values were always higher than 0.95, standard errors of the equation parameters were always low compared to the corresponding parameter values and residues were low and randomly

distributed over the independent variable (not shown). The best fit was obtained using HT, which always had a higher R^2 than the other physiological times, except for 2002, when the best results were obtained with TT (Table 1). The fit of the infection data with HT showed a very close relationship between observed and estimated RII in all the years (Fig. 2).

The use of HT as an independent variable made possible a reduction in between-years variability in estimating RII compared to use of the other physiological times. Using HT, the CV calculated for five important points of the RII curve over the five years were lower than using TT, and especially DH and DD. When RII was 0.05, CV were 41, 47, 75 and 90% for the four models, respectively, and when RII was 0.10, CV were 32, 36, 54 and 63%, respectively. At RII = 0.37 (i.e. the inflection point of the Gompertz equation), 0.5 or 0.9, differences between the five years were low for all the models.

Year	Physiological time ^a		Thermal time	Hydro-thermal time
	Degree-days (/100)	Degree-hours (/1000)		
1999	0.964	0.976	0.976	0.986
2000	0.966	0.955	0.971	0.976
2001	0.971	0.973	0.986	0.992
2002	0.991	0.994	0.996	0.990
2003	0.968	0.965	0.950	0.969
Pooled	0.880	0.854	0.856	0.887

^aPhysiological times were calculated using equations (1), (2), (3) and (4), respectively.

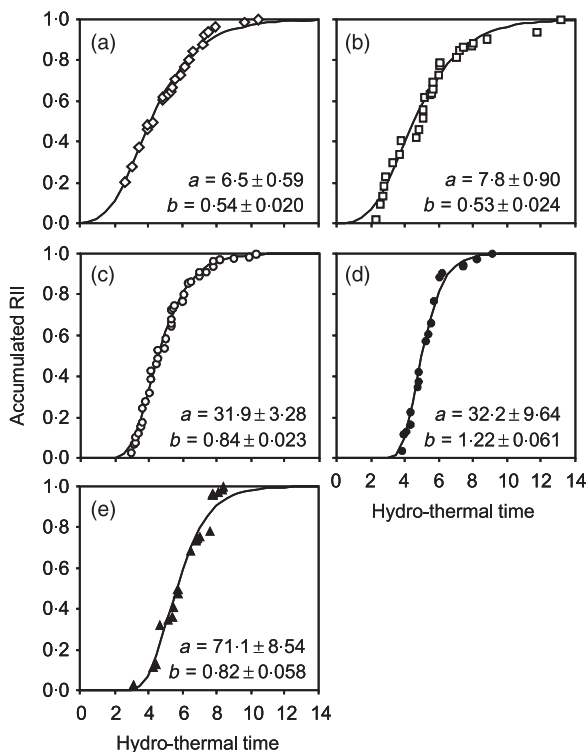


Figure 2 Relationships between accumulated incidence of *Plasmopara viticola* infection in grape leaf discs kept in contact with oospores and the physiological time for oospore germination, calculated as a function of air temperature, rainfall and vapour pressure deficit (hydro-thermal time), yearly from 1999 (a) to 2003 (e). RII is a proportion of the total number of leaf discs infected. Lines show the fit of the Gompertz equation to experimental data: $y = \exp[-a \cdot \exp(-b \cdot x)]$; a and b are the equation parameters with their standard errors.

Gompertz equations calculated using the pooled data for the five years (Table 1) also showed a good accuracy, with R^2 increasing from 0.880 for DD to 0.887 for HT, low standard errors of parameters and a regular distribution of the observed data over the estimated data (Fig. 3). About 43 and 35 out of 124 cases were outside the confidence bands of the calculated curve for DD and DH, respectively, but the outside points diminished for TT (21 cases) and for HT (26 cases); in the HT graph the majority of the outside points was closer to the confidence bands (Fig. 3d) than in the others (Fig. 3a–c).

Table 1 Coefficients of determination (R^2) of the Gompertz equations fitting the relationship between physiological time and relative incidence of infection by *Plasmopara viticola* in grape leaf discs kept in contact with oospores collected in 1999 to 2003, and pooled over the 5 years

The cross-validation showed that HT also performed better than the other physiological times in estimating the RII values of different years. The coefficients of correlation between observed and the cross-estimated data were higher for HT than for DD, DH or TT, with the exception of 2002 (Table 2). The box plots drawn using the differences 'observed minus cross-estimated' showed that the box including 50% of the data was narrower and the whiskers were shorter for HT, particularly for the over-estimates (negative values) (Fig. 4a): the highest deviations were -0.300 , -0.343 , -0.354 and 0.258 with the four methods, respectively. When the HT-cross-estimated data were regressed against the observed data, the intercept and the slope of the regression line were not significantly different from zero and one, respectively, R^2 was 0.894 and SE est. was 0.092 (Fig. 4b).

Hydro-thermal time and field observations

The HT-dependent Gompertz equation was calculated for each year using the weather data collected in the vineyard and the results were compared with the date of grapevine bud break and of the first seasonal *P. viticola* infection.

In 1999 (Fig. 5a), bud break of vines was observed in the last five days of April, when HT was between 2.1 and 2.3; the corresponding proportion of the seasonal oospores calculated with the proper Gompertz equation (Fig. 3d) was less than 3%. The first seasonal disease symptoms appeared on 27 May as a consequence of two possible infection periods, on 15 May, with 7.7 mm of rainfall and 13 h of wetness at 15.3°C, and on 17 May, with 14.6 mm of rainfall and 30 h wetness at 12.7°C. At these times, HT was 3.3 and 3.6, respectively (19 to 21% of the seasonal oospores). In 2000 (Fig. 5b), vines broke dormancy between 20 and 25 April. In this period, HT ranged between 2.2 and 2.4, which is less than 3.5% of the seasonal oospores. The first oil spots appeared on leaves on 18 May, caused by an infection that occurred on 9 to 10 May, when there was 7.2 mm of rainfall followed by 18 h of wetness at 16.3°C. The corresponding HT was 3.4 to 3.6, and the seasonal oospores 18 to 21%. In 2001 (Fig. 5c), the bud break occurred around 25 April, with HT of about 2.8 (less than 8% of the seasonal oospores). The first disease onset occurred on 17 May. The corresponding infection occurred on 6 to 7 May, with 9 mm of rainfall and a wetness period of 16 h at 13.6°C, when HT

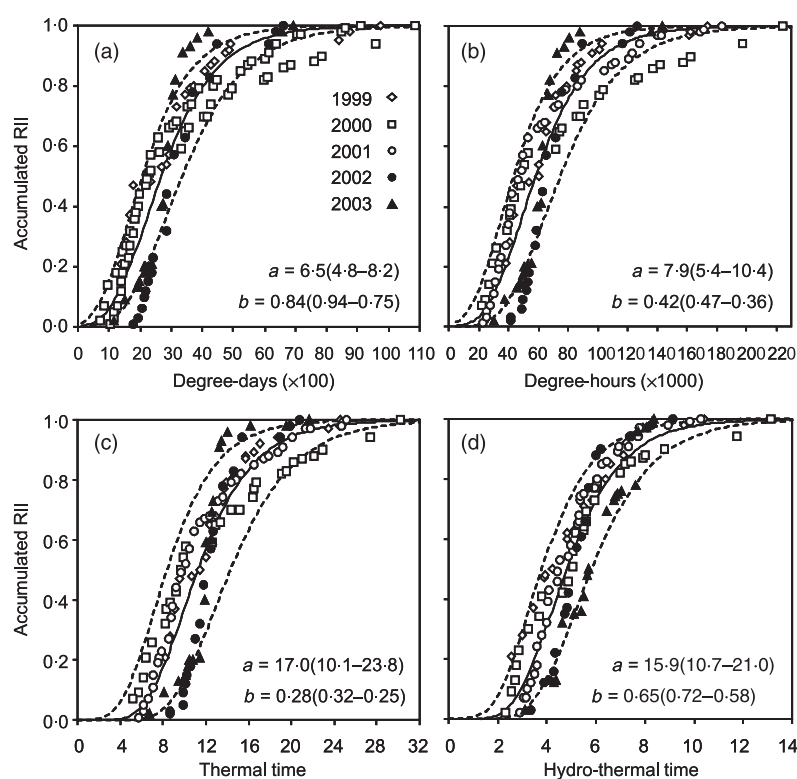


Figure 3 Fit of the relative accumulated incidence of *Plasmopara viticola* infection in grape leaf discs kept in contact with oospores collected from 1999 to 2003 by the equations of Gompertz calculated over the physiological time, expressed as degree-days (a), degree-hours (b), thermal time (c) and hydro-thermal time (d). RII is a proportion of the total number of leaf discs infected. Points represent observed data, lines show the fit (—) and its 95% confidence bands (---) for the pooled data, drawn using the parameters (a and b) and their superior and inferior confidence limits (between brackets): $y = \exp[-a \cdot \exp(-b \cdot x)]$.

Table 2 Pearson's coefficients of correlation between observed and cross-estimated incidences of *Plasmopara viticola* infection in grape leaf discs kept in contact with oospores collected in 1999 to 2003, and coefficients and statistics of the regression lines fitting the pooled data

Year	Physiological time ^a			
	Degree-days (/100)	Degree-hours (/100)	Thermal time	Hydro-thermal time
1999	0.982 ^b	0.984	0.988	0.990
2000	0.975	0.967	0.965	0.984
2001	0.978	0.970	0.979	0.988
2002	0.991	0.983	0.978	0.973
2003	0.945	0.932	0.918	0.962
Pooled	0.916	0.897	0.900	0.929

^aPhysiological times were calculated using equations (1), (2), (3) and (4), respectively.

^bCorrelation coefficients are all significant at $P < 0.0001$, with $n = 22, 28, 36, 18$ and 20 in the five years, respectively.

was 3.5 to 3.6 and the seasonal oospores 19 to 21%. In 2002 (Fig. 5d), vine buds began to vegetate in the first days of May, when HD was 2.8 to 2.9 (7 to 10% of the seasonal oospores). Disease symptoms did not appear until the beginning of June, caused by infections which occurred on 23 May (16.8 mm rainfall, 25 h wetness at 14.4°C) and on 25 May (12.6 mm rainfall, 17 h wetness at 15.1°C). On 23 and 25 May, HT values were 4.1 and 4.2, respectively, with oospore doses of 32 and 37%, respectively. In 2003 (Fig. 5e), bud break was observed in the last days of April (HT of about 2.5, about 4% of the seasonal oospores). Downy mildew did not appear during the season because there was only sporadic rainfall.

Values of HT both when the grapevine buds broke and when the first seasonal downy mildew infections appeared

were almost consistent over the season, ranging between 2.1 and 2.9, and 3.3 and 4.2, respectively.

Discussion

In this work, the germination course of *P. viticola* oospores was determined using the bioassay of Hill (1998). It is based on: (i) sampling an oospore population that has overwintered under natural conditions; (ii) testing its germination capabilities in the laboratory indirectly, based on infection occurrence in grape leaf discs kept in contact with oospores; and (iii) measuring germination as both the time required for infection to occur and the infection frequency. This method produces information on both the age-structure of the oospore population,

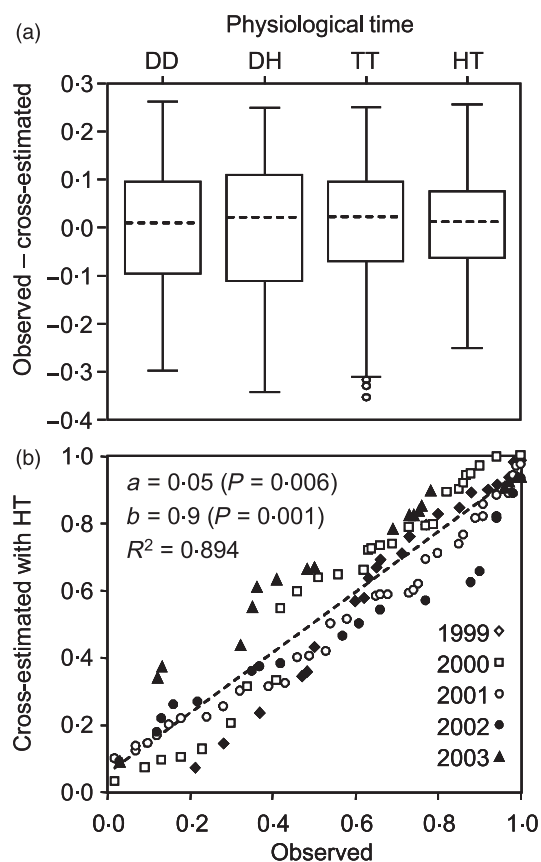


Figure 4 (a) Box and whiskers plots showing distributions of the differences between observed and cross-estimated accumulated incidences of *Plasmopara viticola* infection in grape leaf discs kept in contact with oospores collected from 1999 to 2003, calculated using four physiological times (degree-days, DD; degree-hours, DH; thermal time, TT; hydro-thermal time, HT): boxes include 50% of the data, the dotted line is the median, whiskers extend to minimum and maximum values, and points are outliers. (b) Relationship between observed and cross-estimated data for HT; --- is the regression line; a and b are intercept and slope of the regression, respectively, with their probability level for the null hypotheses that $a = 0$ and $b = 1$. Cross-estimates for a year (for instance 1999) were calculated using the Gompertz equations fitting the data of four years (2000 to 2003) with exclusion of the considered year (1999).

because the same oospore sample remains under optimum conditions for germination to occur for a long time (14 days), and because of the relative abundance of oospores, which is proportional to the infection frequency on the leaf discs.

In previous works, the dynamics of oospore germination were plotted against physical time (e.g. days of the year), while in this work time was measured as physiological time with the aim of accounting for the effect of temperature on the considered phenomenon (Lovell *et al.*, 2004). This approach has been successfully used for fitting data on the development of different overwintering bodies, such as the pseudothecia of *Venturia inaequalis*, the apple scab fungus (James & Sutton, 1982), and of *Pleospora alii*,

the teleomorph of *Stemphylium vesicarium*, the causal agent of brown spot of pear (Montesinos *et al.*, 1995), as well as for maturation of ascospores of *V. inaequalis* (Gadoury & MacHardy, 1982) and *V. pirina* (Spotts & Cervantes, 1994).

In previous work on *P. viticola*, Gehmann (1987) calculated the degree-days between 1 January and the day when the first seasonal germination had occurred, but he did not consider the following course of oospore germination over the season. Therefore, this is the first published work that relates germination dynamics of *P. viticola* oospores to physiological time.

Data on the germination course of the oospores (more precisely the indirect measurements of germination through infection incidence) were regressed against physiological time by means of some mathematical equations. In a preliminary analysis, the equation of Gompertz was better than the others assayed, including the logistic equation, which has been widely used to describe the development over time of various diseases. The equation of Gompertz has a positive asymmetry and an earlier inflection point compared to the logistic, so that the absolute rate approaches the inflection point more rapidly and declines more slowly. This means that germination of *P. viticola* oospores is not symmetrical around the inflection point: it is greater in the early than in the late season. This can be interpreted as an ecological adaptation because in the early season, weather conditions are more favourable for infection: rainfall is more frequent, wet periods are longer and temperatures are closer to the optimal values.

Previously, Park *et al.* (1997) represented the dynamic of oospore maturation over the days of the year by a normal distribution curve. These data cannot be compared with those of the current study because time is expressed in different units: days of the year and physiological time, respectively. Furthermore, it must be considered that the pattern of the physiological time-dependent Gompertz equation changes considerably when plotted against days of the year because physiological time accumulates slowly in the early season as temperatures are low.

The equation of Gompertz has been used to fit the progress of downy mildew epidemics under many different field conditions (Liberati & Vercesi, 1999). Therefore, both oospore germination and downy mildew epidemics follow a similar pattern. This is in agreement with the results obtained with molecular analyses of the population structure of *P. viticola* in natural epidemics which acknowledge a relevant role for oospores in the disease progress for a long time during the grapevine-growing season (Gessler *et al.*, 2003).

Four different methods were used to calculate the physiological time for oospore germination. The first method was degree-days (DD) from Gehmann (1987), which accumulates daily the positive differences between the daily mean temperature and 8°C, starting from 1 January. This was the least successful of the methods used in this work and its results were not in agreement with those of Gehmann (1987). This author found that oospore

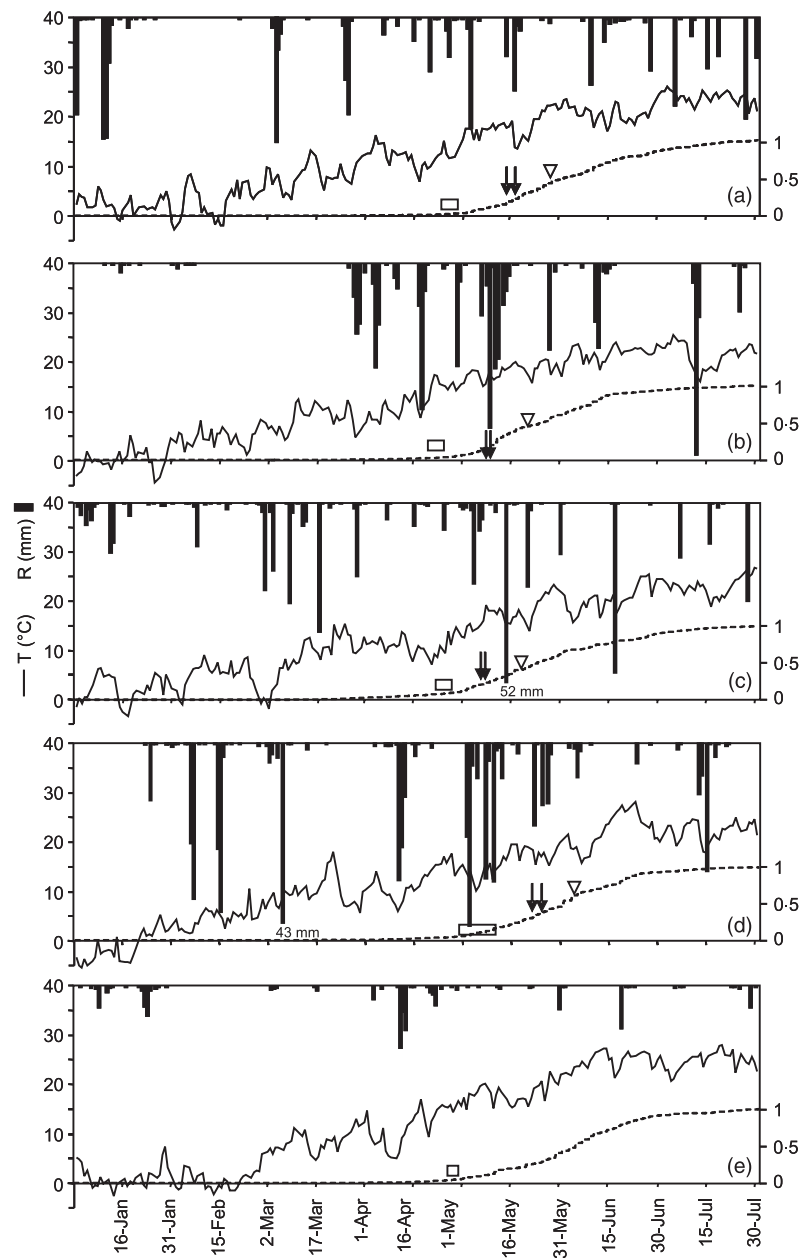


Figure 5 Daily data of air temperature (T , —) and rainfall (R , as 40 minus rain, ■) registered yearly from 1999 (a) to 2003 (e) in the experimental vineyard, and the corresponding dynamics of the hydro-thermal-dependent Gompertz equation (----, in a 0 to 1 scale). This line represents the proportion of the seasonal oospores that should germinate under favourable conditions for germination to occur. Symbols show: time of grapevine bud break (□), first downy mildew onset (▽) and corresponding time of probable infection (↓).

germination started when DD was at least 160, while in 18 cases out of 124, the current study observed germinations with lower DD, with a minimum of about 71.

The second method was a sum of degree-hours (DH), which should account for fluctuations in daily temperatures better than DD (Lovell *et al.*, 2004), with the base temperature of 10°C used by Baldacci (1947) and commonly accepted as the minimum temperature for the processes involved in oospore-derived infections.

The third method, named TT (thermal time), was based on the summation of hourly rates from a temperature-dependent function (Curry & Feldman, 1987). The temperature-dependent function used in this work was

calculated using data from Laviola *et al.* (1986), who incubated oospores at constant temperature regimes in water and measured their germination course.

The fourth method, called hydro-thermal time (HT), was similar to TT, but hourly rates were accumulated only when either rainfall or the water flux from the atmosphere (estimated by VPD) ensured moisture for oospore development (Rossi & Caffi, 2007). The concept of hydro-thermal time has been used for plant growth (Colbach *et al.*, 2005) with emphasis for seed germination and dormancy (Bradford, 2002), but similar methods to calculate the physiological time have also been used in plant pathology (Lovell *et al.*, 2004).

In work described here hydro-thermal time produced the best fit for both separate years and pooled data compared to the other methods, as well as the best accuracy in cross-estimating new data. This method also accounted for a great part of the variability in oospore germination between years and both between and within sampling periods observed in this work. Therefore, the equation of Gompertz calculated over the hydro-thermal time, a physiological time accounting for the effects of both temperature and moisture, produces a consistent modelling of the general relationships between germination of *P. viticola* oospores over the season and weather conditions. Nevertheless, further evaluations will be necessary to determine its robustness.

Based on this model, the development of oospores is a progressive phenomenon through the season; it is faster when the temperature is between 20 and 24°C and there is rainfall, or the leaf litter (or the soil) holding oospores is moist because of rainfall or water flux from the atmosphere. This result is in complete agreement with previous knowledge about (i) the effects of weather on oospore germination, and (ii) the duration of the oosporic season.

Concerning weather, it is known that temperature influences both the time required for oospores to mature and germination rates, with optimum temperatures between 20 and 24°C (Laviola *et al.*, 1986), so that low temperatures in early spring delay maturation (Rouzet & Jacquin, 2003). Rain is necessary for triggering oospore germination (Darpoux, 1943), so that oospores always kept in dry conditions cannot germinate (Burruano *et al.*, 1987). Zachos (1959) found a relationship between germination course and the amount of rainfall oospores receive between December and March. Subsequent studies revealed the role of rainfall distribution rather than its total amount (Tran Manh Sung *et al.*, 1990; Serra & Borgo, 1995; Vercesi *et al.*, 1999), and that long dry periods in spring stop oospore maturation, but the process recovers when rainfall returns (Rouzet & Jacquin, 2003). As a consequence, dry periods in spring delay the time of disease onset (Rossi *et al.*, 2002). Burruano *et al.* (1987) hypothesized that rainfall has an indirect effect *via* soil moisture, and Rossi & Caffi (2007) showed that oospores use the water moistening the leaf litter holding them to develop and that fluxes of water from the atmosphere can also supply a sufficient level of moisture during non-rainy periods.

Concerning the duration of the primary oosporic season, it has been demonstrated that oospore germination goes on for an extended time (Zachos, 1959; Ronzon-Tran Manh Sung & Clerjeau, 1989; Serra & Borgo, 1995; Park *et al.*, 1997) and that there is a continuous input of new oospore-derived genotypes to an epidemic from May to August (Gobbin *et al.*, 2005). This behaviour depends on the population structure of the oospore field population. Oospores form within the affected leaf tissue (Arens, 1929) from July until leaf fall (Galbiati & Longhin, 1984; Gehmann, 1987), and therefore the population entering in winter is not homogeneous (Jermini *et al.*, 2003). In winter, oospores reach morphological maturity (Vercesi

et al., 1999) but are not able to germinate because of dormancy (Galet, 1977). When dormancy has broken oospores can start germination under favourable environmental conditions (Ronzon-Tran Manh Sung & Clerjeau, 1989). The oospore population, then, has an age-physiological structure and germination occurs gradually in the season following their formation (Jermini *et al.*, 2003), with some germinating in the following year (Arens, 1929).

Based on the previous considerations, the hydro-thermal time-based model found in this work accounted for the different processes involved in the development of a non-coeval oospore population, i.e. the breaking of dormancy and germination of different age-physiological cohorts. Similarly, hydro-thermal time has been widely used for modelling germination and seedling emergence of seeds of weed plants (Bradford, 2002). This is a biological phenomenon that has analogies with that studied in this work, because breaking of dormancy in weed seeds is a population-based phenomenon (Benech-Arnold *et al.*, 2000).

It must be underlined that the model elaborated in this work does not represent the amount of germinated oospores in the vineyard at any time of the primary season, but the relative density of seasonal oospores that should produce sporangia when the oospores experience favourable conditions for germination. In fact, it is known that oospore germination is intermittent under natural conditions, and the data collected in this work confirm this knowledge. The method used by Burruano *et al.* (1995) for explaining some primary infections which occurred in Sicily can provide a sound base for (i) a rational understanding of the intermittent nature of oospore germination, and (ii) a possible use of the current model in practice. Any rain event triggers germination of the part of the oospore field population that has broken dormancy at the time of rain; this oospore cohort will produce sporangia some time after, depending on temperature (Laviola *et al.*, 1986) and the availability of water (Rossi & Caffi, 2007), with its relative density depending on physiological time as considered in this work.

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